Example: EKF
Reference: Thrun, Burgard, Fox, "Probabilistic Robotics," MIT Press 2006

The final measurement is illustrated in Figure 7.4d. Here the Markov localization algorithm multiplies the current belief with the perceptual probability $p(z_t \mid x_t)$. At this point, most of the probability mass is focused on the correct pose, and the robot is quite confident of having localized itself. Figure 7.4e illustrates the robot's belief after having moved further down the hallway.

We already noted that Markov localization is independent of the underlying representation of the state space. In fact, Markov localization can be implemented using any of the representations discussed in Chapter 2. We will now consider three different representations and devise practical algorithms that can localize mobile robots in real time. We begin with Kalman filters, which represent beliefs by their first and second moment. We then continue with discrete, grid representations and finally introduce algorithms using particle filters.

7.5 EKF LOCALIZATION

The extended Kalman filter localization algorithm, or EKF localization, is a special case of Markov localization. EKF localization represent beliefs $bel(x_t)$ by their their first and second moment, that is, the mean μ_t and the covariance Σ_t. The basic EKF algorithm was stated in Table 3.3 in Chapter 3.3. EKF localization shall be our first concrete implementation of an EKF in the context of an actual robotics problem.

Our EKF localization algorithm assumes that the map is represented by a collection of features. Thus, at any point in time t, the robot gets to observe a vector of ranges and bearings to nearby features: $z_t = \{z_t^1, z_t^2, \ldots\}$. We begin with a localization algorithm in which all features are uniquely identifiable. The existence of uniquely identifiable features may not be a bad assumption: For example, the Eiffel Tour in Paris is a landmark that is rarely confused with other landmarks, and it is widely visible throughout Paris. The identity of a feature is expressed by set of *correspondence variables*, denoted c_t^i, one for each feature vector z_t^i. Correspondence variables were already discussed in Chapter 6.6. In our first algorithm, the correspondence is assumed to be known. We then progress to a more general version which allows for ambiguity among features. Instead of the correspondence, the robot observes a feature signature, s_t^i. The second, more general version applied a maximum likelihood estimator to estimate the value of the latent correspondence variable, and uses the result of this estimation as ground truth.

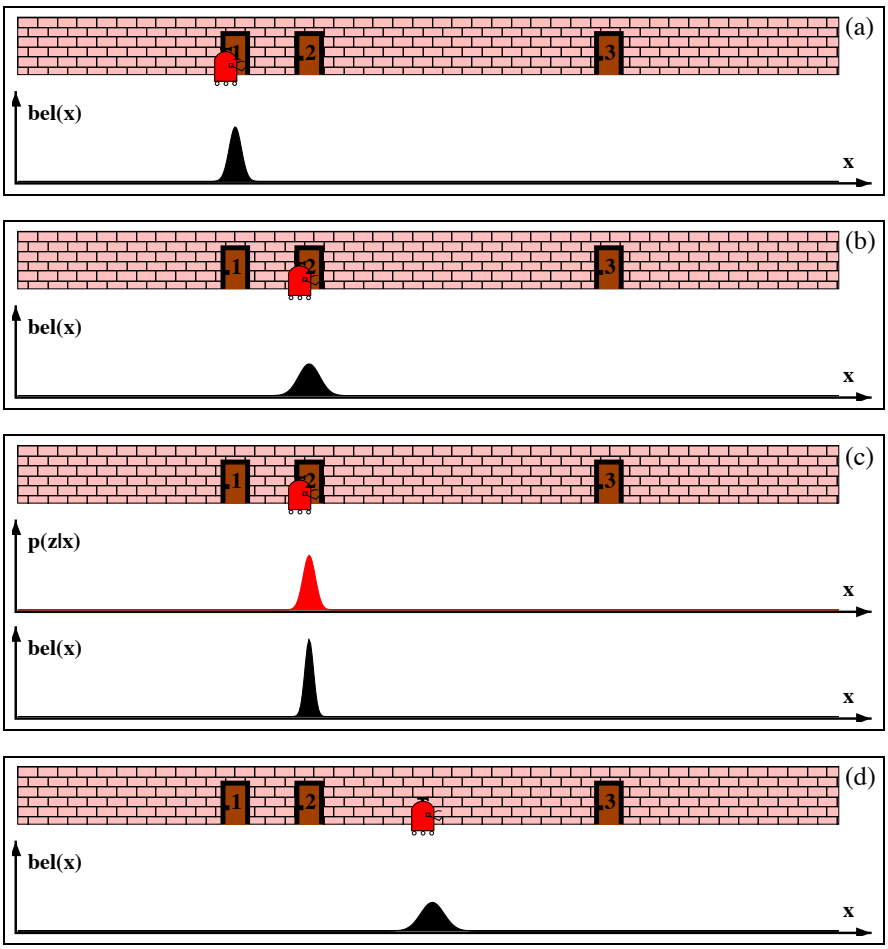


Figure 7.5 Application of the Kalman fi lter algorithm to mobile robot localization. All densities are represented by uni-modal Gaussians.

7.5.1 Illustration

Figure 7.5 illustrates the EKF localization algorithm using our example of mobile robot localization in the one-dimensional corridor environment (*cf*. Figure 7.3). To accommodate the unimodal shape of the belief in EKFs, we make two convenient assumptions: First, we assume that the correspondences are known: we will attach unique labels to each door (1, 2, and 3), and we will denote the measurement model by $p(z_t \mid x_t, c_t)$ where $c_t \in \{1, 2, 3\}$ is the identity of the door observed at time

t. Second, we assume that the initial pose is relatively well known. A typical initial belief is represented by the Gaussian distribution shown in Figure 7.5a, centered on the area near Door 1 and with a Gaussian uncertainty as indicated in that figure. As the robot moves to the right, its belief is convolved with the Gaussian motion model. The resulting belief is a shifted Gaussian of increased width, as shown in Figure 7.5b.

Now suppose the robot detects that it is in front of door $c_t = 2$. The upper density in Figure 7.5c visualizes $p(z_t \mid x_t, m, c_t)$ for this observation—again a Gaussian. Folding this measurement probability into the robot's belief yields the posterior shown in Figure 7.5c. Note that the variance of the resulting belief is smaller than the variances of both the robot's previous belief and the observation density. This is natural, since integrating two independent estimates should make the robot more certain than each estimate in isolation. After moving down the hallway, the robot's uncertainty in its position increases again, since the EKF continues to incorporate motion uncertainty into the robot's belief. Figure 7.5d shows one of these beliefs. This example illustrates the EKF in our limited setting.

7.5.2 The EKF Localization Algorithm

The discussion thus far has been fairly abstract: We have silently assumed the availability of an appropriate motion and measurement model, and have left unspecified a number of key variables in the EKF update. We will now discuss a concrete implementation of the EKF, for feature-based maps. Our feature-based maps consist of point landmarks, as already discussed in Chapter 6.2. For such point landmarks, we will use the common measurement model discussed in Chapter 6.6. We will also adopt the velocity motion model defined in Chapter 5.3. The reader may take a moment to briefly reacquire the basic measurement and motion equations discussed in these chapters before reading on.

Table 7.2 describes **EKF_localization_known_correspondences**, the EKF algorithm for localization with known correspondences. This algorithm is derived from the EKF in Table 3.3 in Chapter 3. It requires as its input a Gaussian estimate of the robot pose at time $t-1$, with mean μ_{t-1} and covariance Σ_{t-1}. Further, it requires a control u_t, a map m, and a set of features $z_t = \{z_t^1, z_t^2, \ldots\}$ measured at time t, along with the correspondence variables $c_t = \{c_t^1, c_t^2, \ldots\}$. It output is a new, revised estimate μ_t, Σ_t.

The individual calculations in this algorithm will be explained further below. Lines 2 through 4 implement the familiar motion update, using a linearized motion model. The predicted pose after motion is calculated as $\bar{\mu}_t$ in Line 2, and Line 4 computes the

1: **Algorithm EKF_localization_known_correspondences**($\mu_{t-1}, \Sigma_{t-1}, u_t, z_t, c_t, m$)**:**

2: $\bar{\mu}_t = \mu_{t-1} + \begin{pmatrix} -\frac{v_t}{\omega_t}\sin\mu_{t-1,\theta} + \frac{v_t}{\omega_t}\sin(\mu_{t-1,\theta} + \omega_t\Delta t) \\ \frac{v_t}{\omega_t}\cos\mu_{t-1,\theta} - \frac{v_t}{\omega_t}\cos(\mu_{t-1,\theta} + \omega_t\Delta t) \\ \omega_t\Delta t \end{pmatrix}$

3: $G_t = \begin{pmatrix} 1 & 0 & \frac{v_t}{\omega_t}\cos\mu_{t-1,\theta} - \frac{v_t}{\omega_t}\cos(\mu_{t-1,\theta} + \omega_t\Delta t) \\ 0 & 1 & \frac{v_t}{\omega_t}\sin\mu_{t-1,\theta} - \frac{v_t}{\omega_t}\sin(\mu_{t-1,\theta} + \omega_t\Delta t) \\ 0 & 0 & 1 \end{pmatrix}$

4: $\bar{\Sigma}_t = G_t\ \Sigma_{t-1}\ G_t^T + R_t$

5: $Q_t = \begin{pmatrix} \sigma_r & 0 & 0 \\ 0 & \sigma_\phi & 0 \\ 0 & 0 & \sigma_s \end{pmatrix}$

6: *for all observed features* $z_t^i = (r_t^i\ \ \phi_t^i\ \ s_t^i)^T$ *do*

7: $j = c_t^i$

8: $\delta = \begin{pmatrix} \delta_x \\ \delta_y \end{pmatrix} = \begin{pmatrix} m_{j,x} - \bar{\mu}_{t,x} \\ m_{j,y} - \bar{\mu}_{t,y} \end{pmatrix}$

9: $q = \delta^T\delta$

10: $\hat{z}_t^i = \begin{pmatrix} \sqrt{q} \\ \mathrm{atan2}(\delta_y, \delta_x) - \bar{\mu}_{t,\theta} \\ m_{j,s} \end{pmatrix}$

11: $H_t^i = \frac{1}{q}\begin{pmatrix} \sqrt{q}\delta_x & -\sqrt{q}\delta_y & 0 \\ \delta_y & \delta_x & -1 \\ 0 & 0 & 0 \end{pmatrix}$

12: $K_t^i = \bar{\Sigma}_t\ H_t^{i,T}(H_t^i\ \bar{\Sigma}_t\ H_t^{i,T} + Q_t)^{-1}$

13: *endfor*

14: $\mu_t = \bar{\mu}_t + \sum_i K_t^i(z_t^i - \hat{z}_t^i)$

15: $\Sigma_t = (I - \sum_i K_t^i\ H_t^i)\ \bar{\Sigma}_t$

16: *return* μ_t, Σ_t

Table 7.2 The extended Kalman filter (EKF) localization algorithm, formulated here for a feature-based map and a robot equipped with sensors for measuring range and bearing. This version assumes knowledge of the exact correspondences.

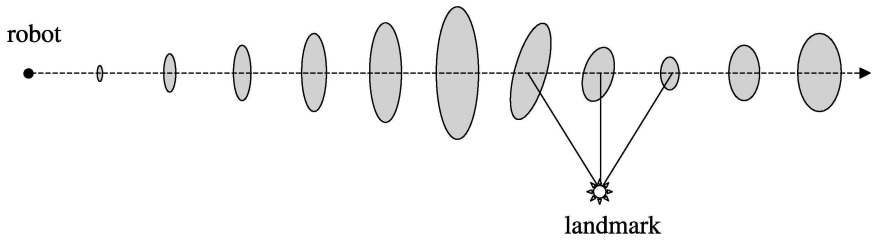


Figure 7.6 Example of localization using the extended Kalman filter. The robot moves on a straight line. As it progresses, its uncertainty increases gradually, as illustrated by the error ellipses. When it observes a landmark with known position, the uncertainty is reduced.

corresponding uncertainty ellipse. Lines 5 to 15 implement the measurement update. The core of this update is a loop through all possible features i observed at time t. In Line 7, the algorithm assigns to j the correspondence of the i-th feature in the measurement vector. It then calculates a predicted measurement $\hat{z}_t^i$ and the Jacobian H_t^i of the measurement model. The Kalman gain K_t^i is then calculated in Line 12 for each observed feature. The sum of all updates is then applied to obtain the new pose estimate, as stated in Lines 14 and 15. Notice that the last row of H_t^i is all zero. This is because the signature does not depend on the robot pose. The effect of this degeneracy is that the observed signature s_t^i has no effect on the result of the EKF update. This should come at no surprise: knowledge of the correct correspondence z_t^i renders the observed signature entirely uninformative.

Figure 7.6 illustrates the EKF localization algorithm in a synthetic environment with a single landmark. The robot starts out on the left and accrues uncertainty as it moves. Upon seeing the landmark, its uncertainty is gradually reduced as indicated.

7.5.3 Mathematical Derivation

To understand the motion update, let us briefly restate the motion model that was defined in Equation (5.13):

$$\begin{pmatrix} x' \\ y' \\ \theta' \end{pmatrix} = \begin{pmatrix} x \\ y \\ \theta \end{pmatrix} + \begin{pmatrix} -\frac{v_t}{\omega_t}\sin\theta + \frac{v_t}{\omega_t}\sin(\theta + \omega_t\Delta t) \\ \frac{v_t}{\omega_t}\cos\theta - \frac{v_t}{\omega_t}\cos(\theta + \omega_t\Delta t) \\ \omega_t\Delta t + \gamma_t\Delta t \end{pmatrix} \tag{7.4}$$